



MINING DEALER BEST PRACTICE

D8T-D9T Carrier Shaft Removing Plate

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

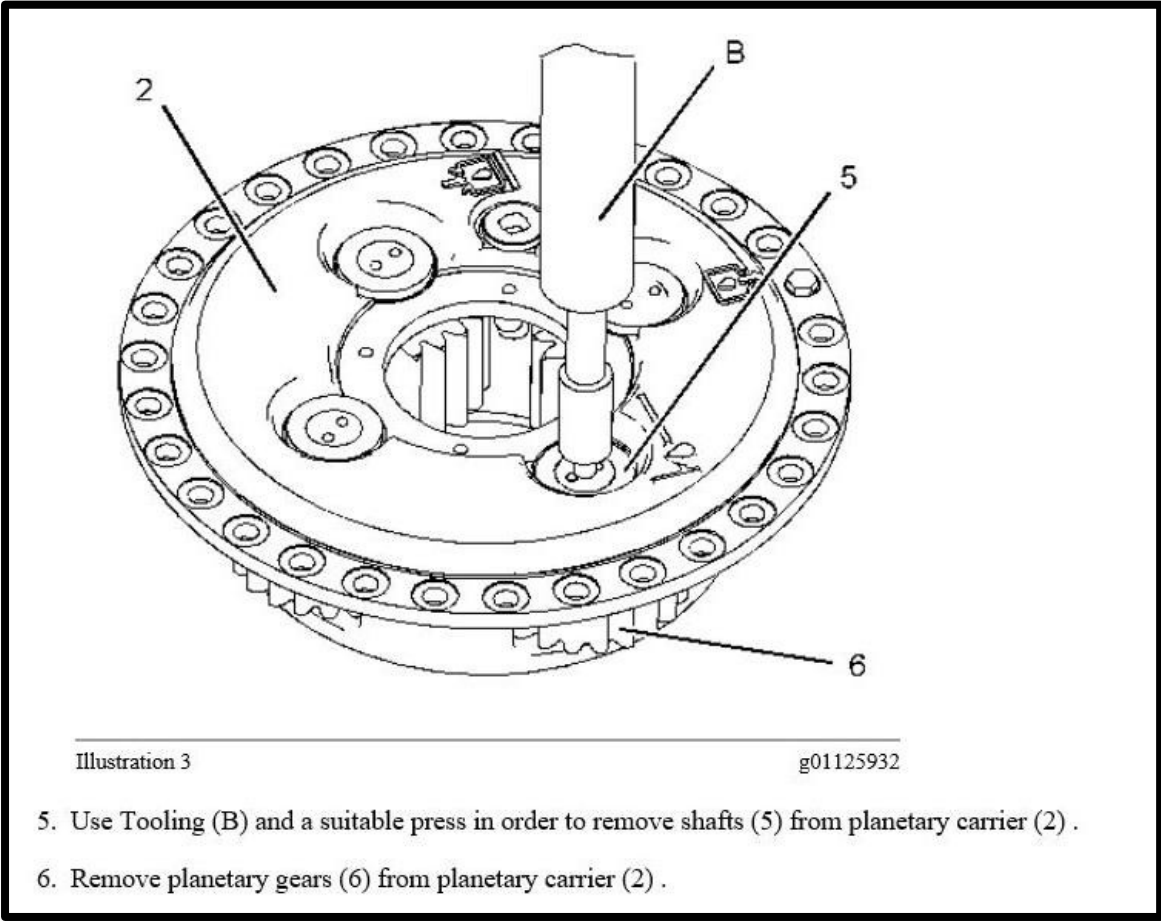
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4.03-210316-2

1.0 Introduction

The final drive disassembly process includes disassembly of the planetary carrier. In the illustration below, which provides information from CAT Service Information System, the fifth step of the process says the shaft should be removed from the planetary carrier. The shaft should be removed from the planetary carrier by using tool B and a suitable press.



However, the underside of the planetary carrier, which is just below the planetary gears, is very thin and might get scratched and damaged when the press is applied. The planetary carrier doesn't stay stable on the ground and kicks back when the press is applied. This might not sound like an important issue at first glance, but it can be very harmful for the technician.

To find a solution for this problem, one of the Powertrain and Hydraulic Service Engineers at Borusan Makina ve Güç Sistemleri has designed a plate that gives support to the underside of the planetary carrier. Thanks to this plate, the planetary carrier doesn't kick back and is protected from any kind of damage.

There is a video in the attachments in which you can see the kick back of the planetary carrier when the plate is not used. Please watch it with sound on and give special attention to seconds four through six. The name of the attachment is "Planetary Carrier_Kicking Back".

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2.0 Best Practice Description

The plate is functional and has holes in it just under the planetary gears, so that it doesn't receive any force from the press when the press is applied to the shaft. The plate is almost twice as thick as the underside of the planetary carrier, so that it keeps the planetary carrier from kicking back. Steel (St-52) was used to fabricate the plates.

Below is the manufactured plate. In the attachments you can find engineering prints for both a three hole plate, for the D9T, and the four hole plate, for the D8T. Implementation of the plate into the disassembly process is very easy.



3.0 Implementation Steps

- 1. Place the plate under the planetary carrier and apply the suitable press. Pay special attention to the hole alignment; the holes should be located just under the planetary gears.

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4.0 Benefits

- Protects planetary carrier from damages such as scratching and breaks;
- Increases customer satisfaction;
- Reduces the risk of occupational-work accident for the technician.

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5.0 Resources Required

Item	Cost
Material	\$60
Labor	\$250
Total	\$310

6.0 Supporting Attachments / References

Planetary Carrier_Kicking Back.mp4
D8T-D9T Carrier Shaft Removing Plate_Engineering Print_3hole.pdf
D8T-D9T Carrier Shaft Removing Plate_Engineering Print_4hole.pdf

7.0 Related Best Practices

N/A.

8.0 Acknowledgements

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